

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 1: Introduction and Review of the Classical Linear Regression Model

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Outline

Course Syllabus

Review of the Classical Linear Regression Model

The Course

Course Coordinator: Fu Ouyang

- Office: Room 532A, Colin Clark Building (39)
- Email: f.ouyang@uq.edu.au
- Consultation Hours: Monday 17:00-19:00 or by appointment

Tutors:

- ECON 3350: Dominic, Michelle, Mitchell, Ransi, and Farzana
- ECON 7350: Pablo and Jesse
- Consultation timetables will be available on SI-net.

Lecture Slides, Recordings, Tutorials and Assessment materials on Blackboard

- Be sure the visit the Blackboard site regularly for updates, announcements, etc.

The Course

Tutorials using Stata

The textbooks:

- Enders, W., *Applied Econometric Time Series*, 4th Edition, Wiley, 2015.
- Hill, R.C., Griffiths, W.E. and G.C. Lim, *Principles of Econometrics*, 4th edition, New Jersey: Wiley, 2011.
- Verbeek, M. J. C. M., *A Guide to Modern Econometrics*, 3rd edition. Chichester: John Wiley and Sons, 2008.

Lectures: Concepts, methods, and examples

Tutorials: Consolidating concepts and applying methods using data
(attending tutorials is important for getting hands-on experience)

Assessment tasks: Different for ECON3350 and 7350; please refer to Blackboard

Assignment Task	Due Date	Weight
Assignment 1	18 Mar 2019 23:59	10%
Mid Semester Exam	8 Apr 2019 16:00-17:10	20%
Assignment 2	6 May 2019 23:59	10%
Final Exam	Examination Period	60%

ECON 7350 Assessment

Assignment Task	Due Date	Weight
Research Report 1	9 Apr 2019 23:59	20%
Research Report 2	28 Jun 2019 23:59	20%
Final Exam	Examination Period	60%

Outline

Course Syllabus

Review of the Classical Linear Regression Model

Review of Classical Regression

Recommended readings

Author	Title	Chapter	Call No
Verbeek	A Guide to Modern Econometrics	2, 3, 4, 5	HB139 .V465 2012
Hill et al.	Principles of Econometrics	1-10	HB139 .H548 2011

The Classical Regression Model

A rudimentary approach in analyzing economic data is to specify a linear relationship:

$$y_i = \beta_1 + \beta_2 x_{2,i} + \cdots + \beta_k x_{k,i} + \varepsilon_i.$$

We refer to y_i as the *dependent* variable, $x_{2,i}, \dots, x_{k,i}$ as the *regressors*, ε_i as the *error term* and $\beta_1, \dots, \beta_k$ as the parameters.

Write this more concisely in matrix notation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ are $n \times 1$ vectors, $\mathbf{X}$ is a $n \times k$ matrix, and $\boldsymbol{\beta}$ is a $k \times 1$ vector.

Given data $\mathbf{y}$ and $\mathbf{X}$, how can we analyze the relationship between the dependent variable and its regressors?

Assumptions of the Classical Regression

To make this setup useful in practice, we need some assumptions. CR has 5 basic assumptions:

- (A1) (Linearity) $y_i = \beta_1 + \beta_2 x_{2,i} + \cdots + \beta_k x_{k,i} + \varepsilon_i$.
- (A2) (Exogeneity) $E[\varepsilon_i | x_i] = 0$, where $x_i = (1, x_{2,i}, \dots, x_{k,i})$.
- (A3) (Homoskedasticity) $Var(\varepsilon_i | x_i) = \sigma^2$.
- (A4) (Random sample) $Cov(\varepsilon_i, \varepsilon_j | x_i, x_j) = 0$ for $i \neq j$.
- (A5) (No perfect multicollinearity) There is no exact linear relationship among the elements of x_i .

Estimation

A sensible approach to estimate β is to minimize the *sum of squared errors*:

$$\hat{\beta} = \underset{b}{\operatorname{argmin}} (\mathbf{y} - \mathbf{X}b)'(\mathbf{y} - \mathbf{X}b).$$

This leads to the *ordinary least squares* (OLS) estimator:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

- As long as Assumption A5 is satisfied, OLS can be computed.
- Most statistical computation packages (Stata, Eviews, R, etc.) can compute OLS even with very large n and k almost instantaneously.

Given Assumptions A1-A5, OLS is the **B**est **L**inear **U**nbiased **E**stimator.

Mean Square Error

Is BLUE necessarily the best we can do?

When choosing an estimator, a typical goal is to minimize:

$$\begin{aligned}\text{MSE} &= \text{tr } E \left((\hat{\beta} - \beta)(\hat{\beta} - \beta)' \right) \\ &= \text{tr } \text{Var}(\hat{\beta}) + \text{tr} \left(E(\hat{\beta}) - \beta \right) \left(E(\hat{\beta}) - \beta \right)'.\end{aligned}$$

For all *unbiased* estimators, $E(\hat{\beta}) = \beta$ and MSE is equivalent to the variance. But we can get lower MSE by trading off variance for bias.

Do we really need *unbiased* estimates?

Unbiased vs. Consistent

An *unbiased* estimator gets it right “on average” in small samples.

Instead, we might seek a *consistent* estimator: one that *converges* (in probability) to the true value with the sample size: $\hat{\beta} \xrightarrow{p} \beta$.

An unbiased estimator may not be consistent, and a consistent estimator can be biased.

- Hence, we can achieve lower MSE, but still ensure that the estimator “gets it right” if the sample is “large enough”.

The consistent estimator that guarantees lowest MSE is the *maximum likelihood estimator* (MLE).

- turns out to be equivalent to OLS under certain conditions

Regression Example

Suppose we want to analyze the relationship between wages, years of experience, and years of schooling *in the population*.

We specify the model:

$$\text{wage}_i = \beta_1 + \beta_2 \text{YOE}_i + \beta_3 \text{YOS}_i + \varepsilon.$$

We assume that $(\text{wage}_i, \text{YOE}_i, \text{YOS}_i)$ have a tri-variate distribution. The regression can help us draw inference about the *conditional expectation*

$$E(\text{wage}_i | \text{YOE}_i, \text{YOS}_i) = \beta_1 + \beta_2 \text{YOE}_i + \beta_3 \text{YOS}_i.$$

Causal inference from the CR

What about the question: how does an additional year of schooling affect expected earnings?

- Have to be very careful about causal inference!
- The derivative $\partial E(\text{wage}_i \mid \text{YOE}_i, \text{YOS}_i) / \partial \text{YOS}_i$ can be ascertained as long as YOS_i does not affect the other observed regressors (i.e. YOE_i) nor any unobserved regressors (jointly hidden in the error term).
- In general, CR makes no claims about the direction of causality. If the assumption are fulfilled, then β_3 is consistently estimated, but if we switch wage_i with YOS_i in the regression, we will still get consistent estimates!

Inference from the CR with Hypothesis Tests

In the classical context, inference is obtained through *hypothesis tests*.

Under mild regularity conditions, we can derive the asymptotic distribution of $\hat{\beta}$.

Use the asymptotic distribution to test relationships, e.g.

- $H_0 : \beta_3 = 0, H_1 : \beta_3 \neq 0$ – rejecting H_0 provides statistical evidence that $wage_i$ is correlated with YOS_i .
- $H_0 : \beta_2 = \beta_3 = 0, H_1 : \beta_2 \neq 0$ or $\beta_3 \neq 0$ – rejecting H_0 provides evidence that $wage_i$ is correlated with at least one of the regressors.

Keep in mind that:

- A **null hypothesis** is **never accepted** (fail to reject means not enough statistical evidence to reject H_0 at the given *significance level*).
- For a fixed *significance level*, we may not be able to reject H_0 even if it is false when the sample size is not large enough.

Choice of Regressors

The right choice of regressors is critical, but extremely difficult.

- This is an important issue in applied macro—we will return to it.

Some guidelines to consider:

- **Never discard** regressors due to **lack of significance**.
- **Do not** rely on **goodness-of-fit** measures such as R^2 , which never decreases as more regressors are added. Consider adjusted R^2 , AIC/BIC.
- Omitting a relevant regressor and including an irrelevant regressor.
- Good approach: try several specifications; use *information criteria* to compare them; report and interpret results across all specifications.

Application to Macro and Finance

In macroeconomics and finance, most data is time-series. For most applications, it is unlikely to have Assumptions A2-A4 all satisfied.

Use CR as a starting point, but develop more suitable methods.

Next week, start discussing [univariate time-series](#).